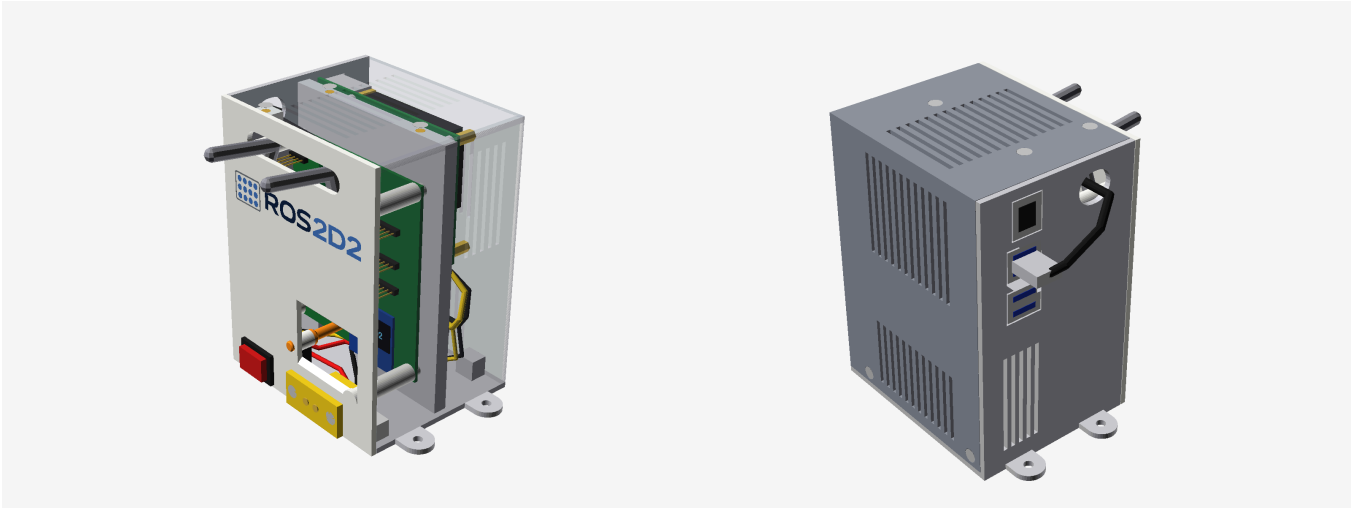
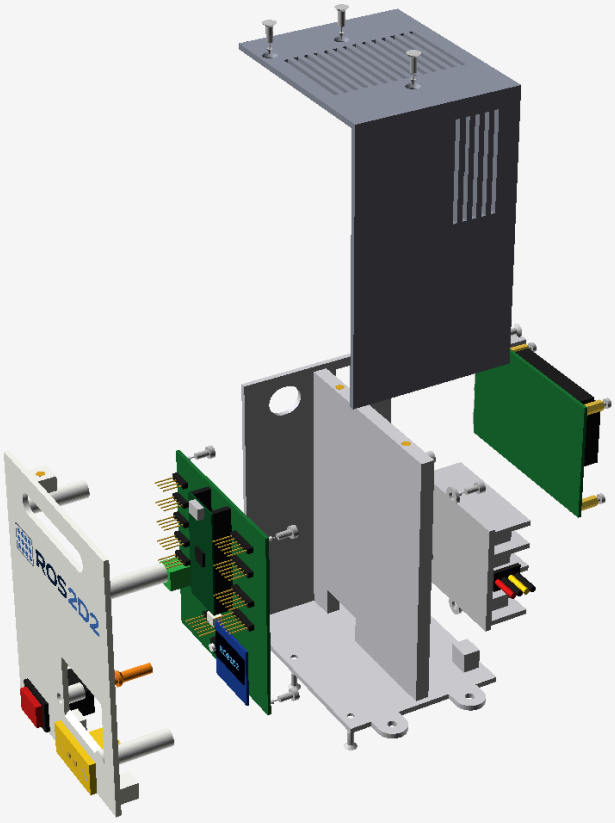
ROS2 rack box · Assembly instructions

Second box, rack style: the Pico board, the Pi and the converter stand on edge, side by side. 78 × 95 × 127 mm (0.94 L, about half the mini tower). OLED, logo, switch and XT60 are on the main face; the Pi's USB at the back. Before printing, measure what the README lists.



Exploded view


ROS202

Hardware and parts

Printed parts

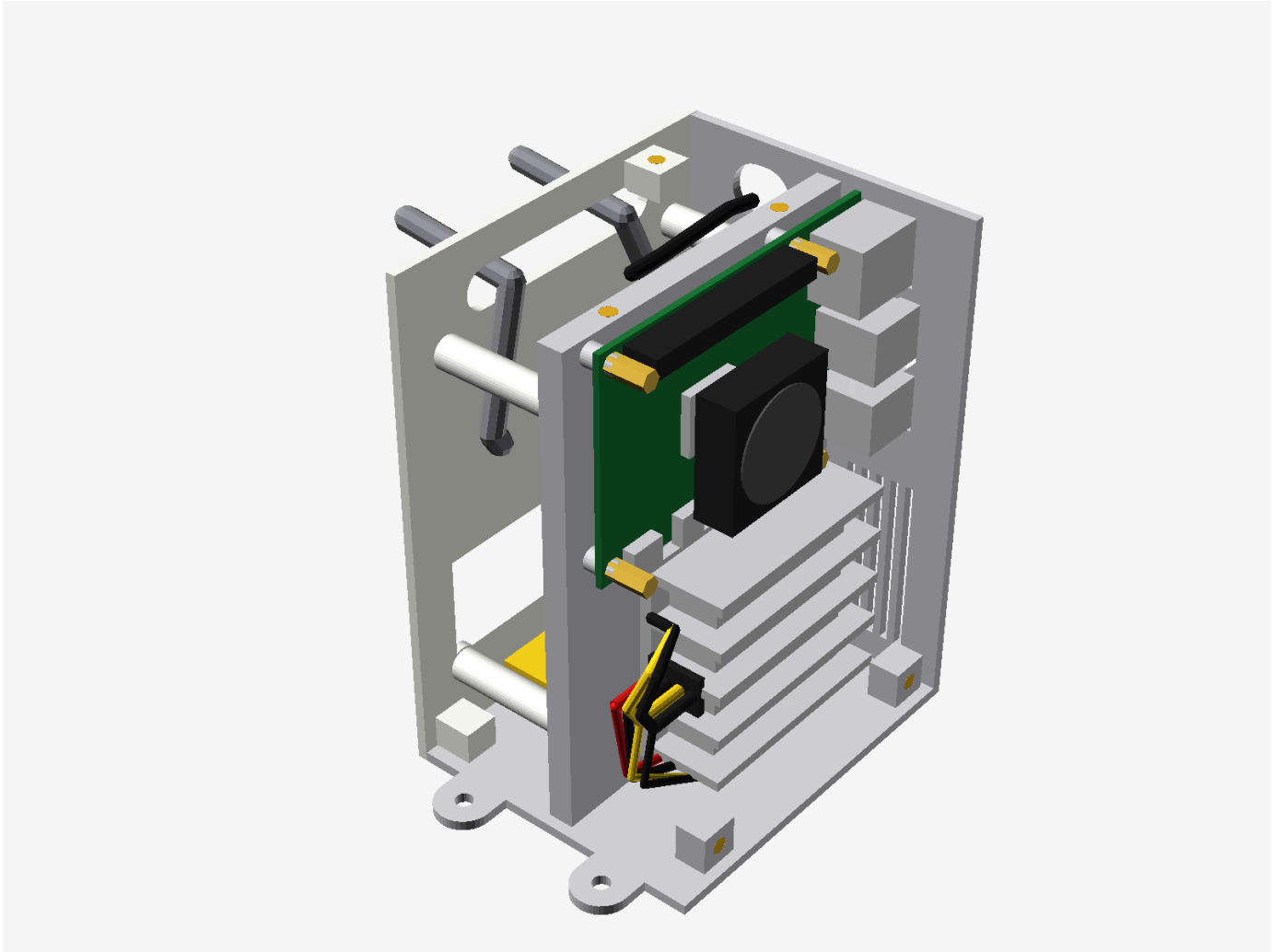
`pared_multicolor_bambu.3mf`	Main face with the logo in 4 colours (AMS: white, navy, blue, grey)
`pared_monocolor.stl`	or: single-colour main face, engraved logo
`base.stl`	Base: floor, back, central divider
`tapa.stl`	Shell (roof + side + narrow front)
`pulsador.stl`	DISPLAY push rod

Qty	Part	Step
13×	M3 heat-set insert (5.7 mm long, 4.0 mm hole)	1
7×	M3 × 8 countersunk screw (ISO 10642)	6, 7
4×	M3 × 6 button head screw (ISO 7380)	5
2×	M3 × 6 socket head cap screw (DIN 912)	2
2×	M3 × 10 countersunk screw (ISO 10642)	4
2×	M3 nut (DIN 934)	4
4×	M2.5 × 6 screw (longer if the Pi's fan sits on spacers)	3
4×	M4 × 12 screw (or what the R2's mount needs)	8

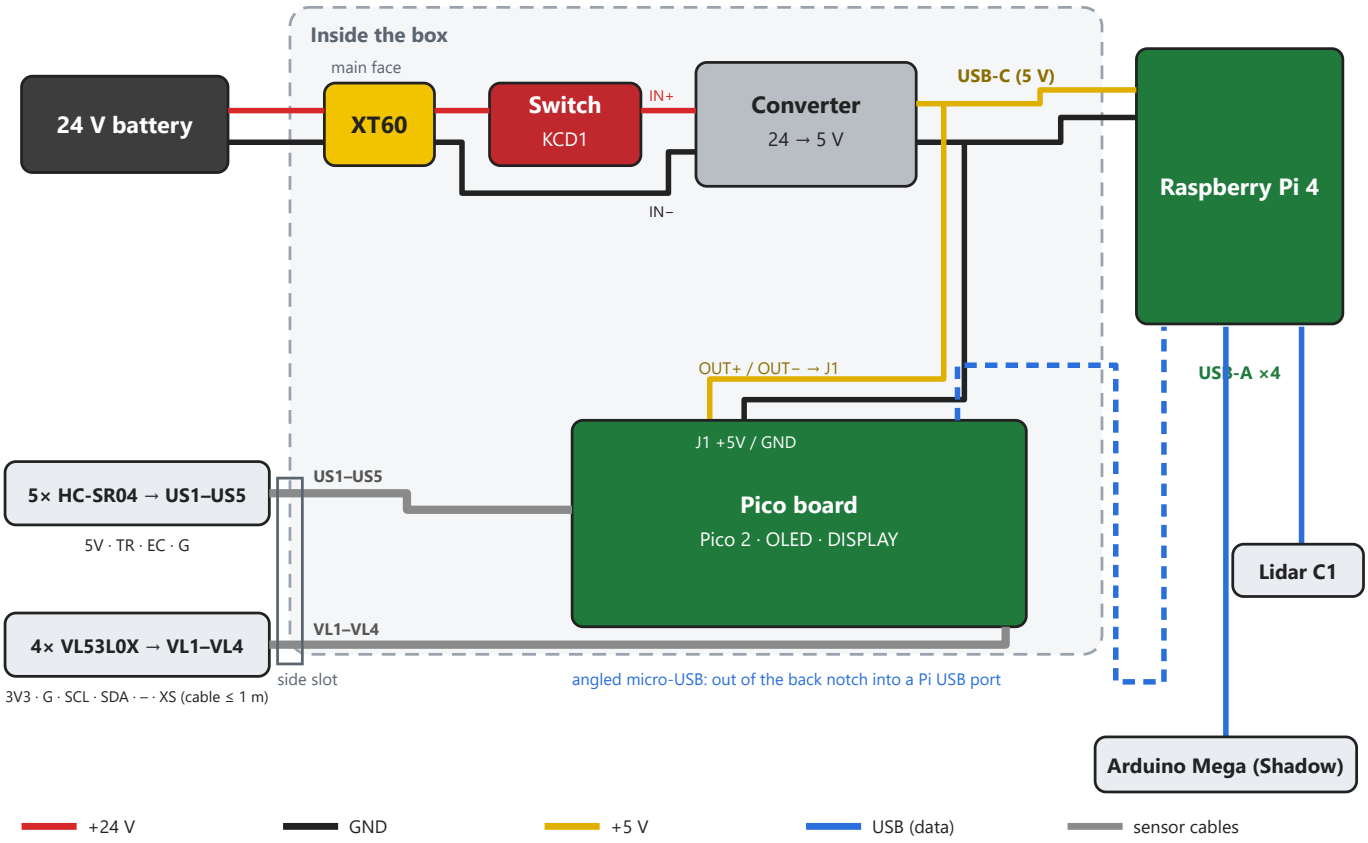
Other parts

1	KCD1 rocker switch (panel hole 19.2 × 13 mm)
1	XT60E-F panel connector (AMASS)
1	micro-USB 2.0 to USB-A cable, 90° side angle, 30 cm (see «Angled plugs»)
1	USB-C male pigtail (5 V to the Pi), 3 A, 15–20 cm; straight or 90° angled
—	0.5 mm^2 wire for the 5 V to terminal J1, 1 mm^2 for the 24 V
1	Edge grommet strip for the 35 × 10 mm slot

Inside (no shell)



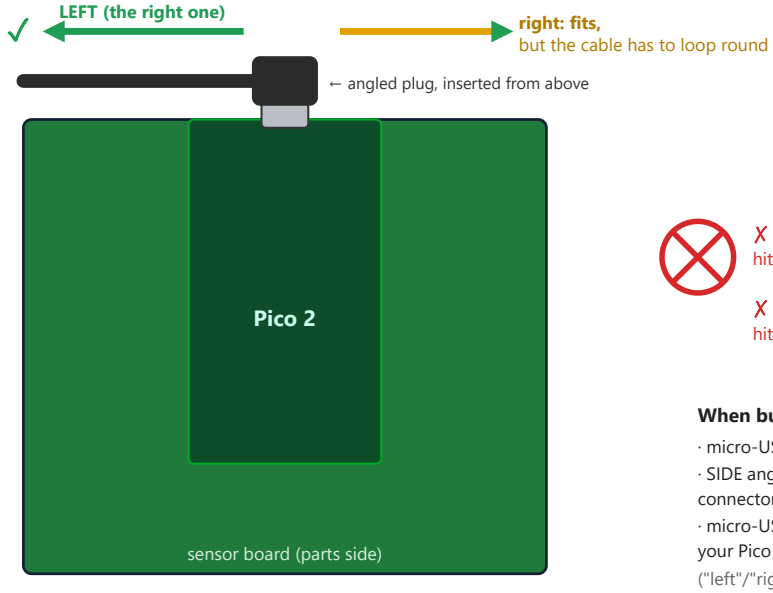
Wiring



Angled plugs: which way the cable goes

1 · Angled micro-USB (Pico → Pi) · both boxes

Looking at the Pico's parts side, its USB at the top (as it sits on the sensor board):



✗ **towards you (the parts side):**
hits the sensors' Dupont plugs

✗ **backwards (behind the board):**
hits the divider in box C

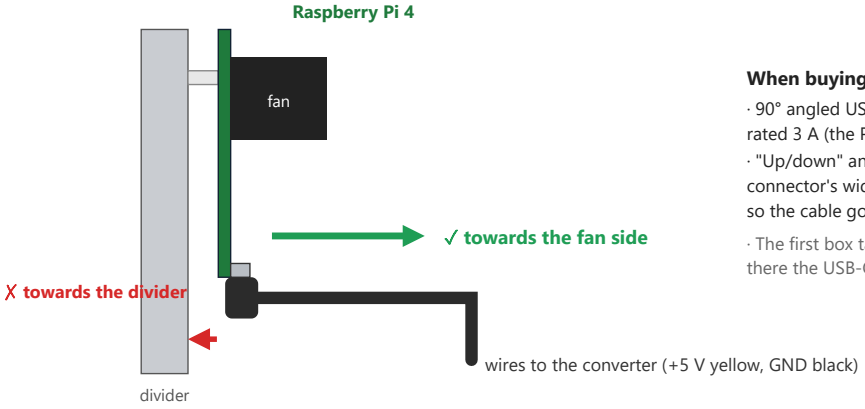
When buying:

· micro-USB 2.0 (not 3.0), 30 cm (fits both boxes).
· SIDE angle: the cable leaves sideways, along the connector's long side (not up or down).
· micro-USB only goes in one way: compare the photo with your Pico and check the cable will end up on the LEFT.
("left"/"right" mean different things to different brands)

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2 · Angled USB-C pigtail (converter → Pi) · box C

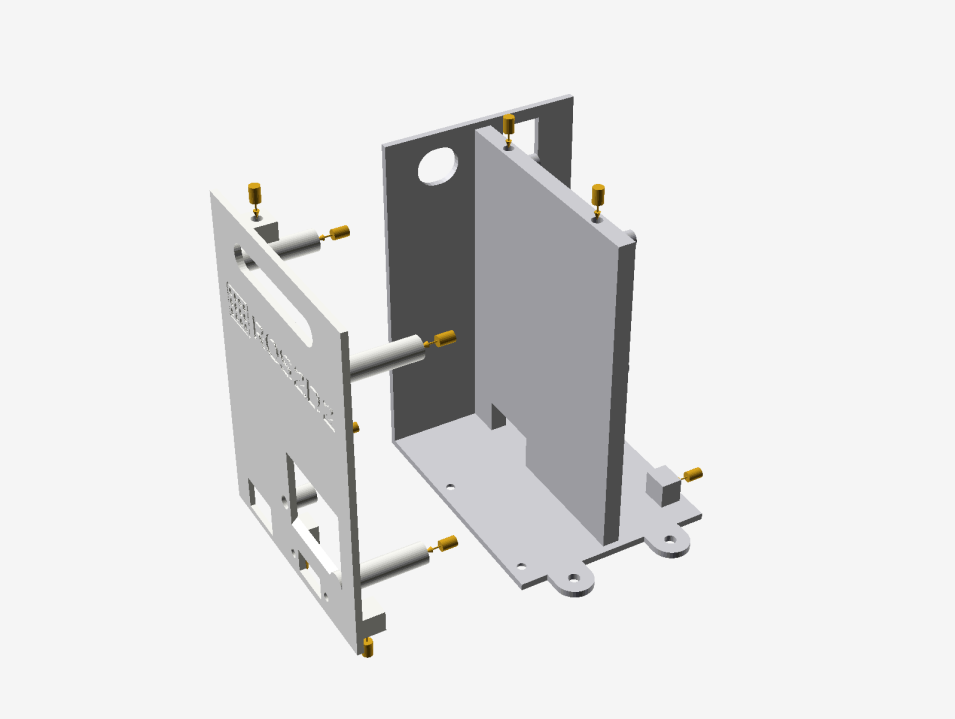
The Pi seen from the narrow front: standing, parts and fan to the right, USB-C at the bottom:



When buying:

· 90° angled USB-C male pigtail (loose wires), rated 3 A (the Pi 4 wants 5 V / 3 A), 15–20 cm.
· "Up/down" angle: the cable leaves square to the connector's wide face. USB-C is reversible: turn it so the cable goes towards the fan.
· The first box takes the same cable (or a straight one): there the USB-C faces the roof, cable towards the middle.

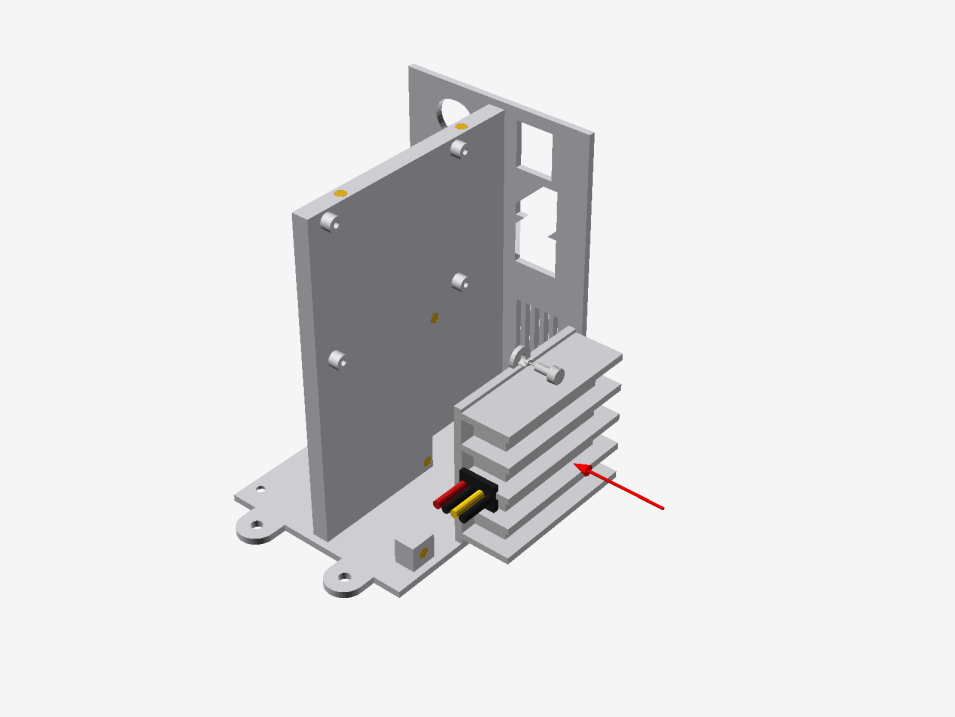
1 Heat-set inserts



13× M3 insert

With the iron at about 220 °C press the 13 inserts in straight and flush: on the main face, 4 in the columns, 2 in the bottom tabs and 1 in the top tab; on the base, 2 on top of the divider, 2 in the divider for the converter and 2 in the floor blocks.

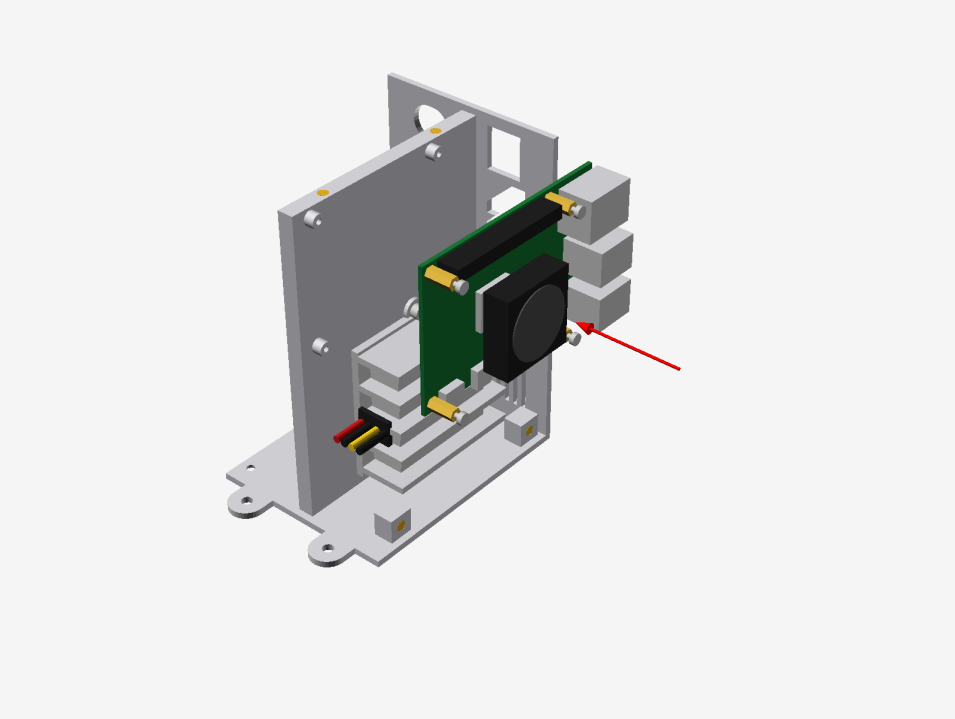
2 Converter onto the divider



2× M3 × 6 socket head

On edge, its base plate against the divider, fins outwards, wires to the front. Check its holes (`conv_holes`) with a caliper first.

3 Raspberry Pi onto the divider



4× M2.5 × 6

On its 4 standoffs, USB and Ethernet flush with the back, USB-C at the bottom. Plug in the angled USB-C and join it to the converter's 5 V output.

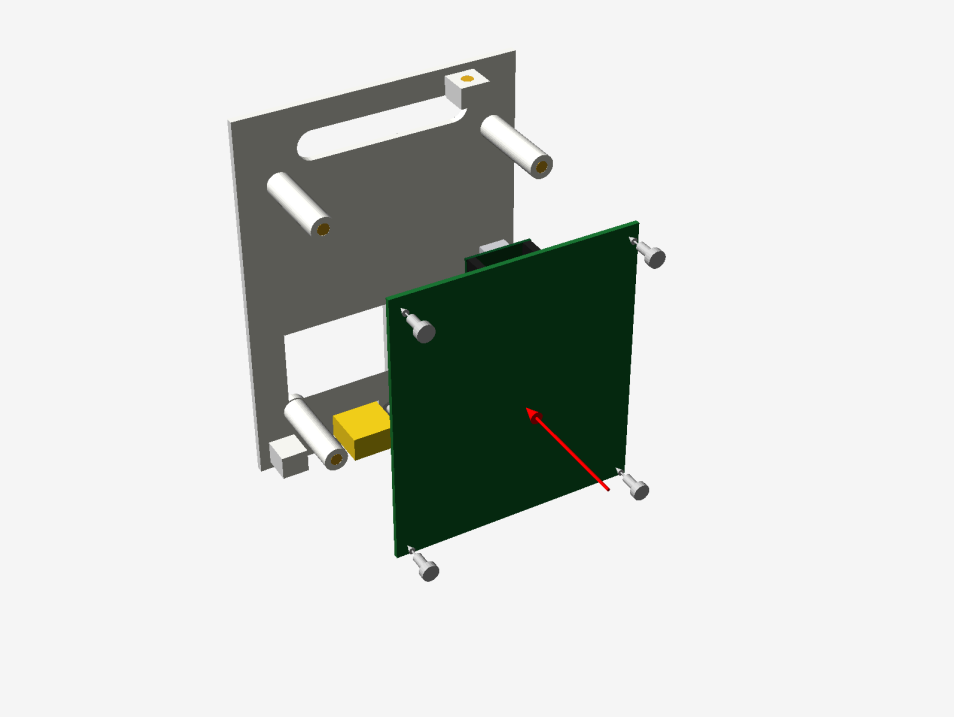
4 XT60 and switch on the main face



2× M3 × 10 countersunk | 2× M3 nut

From outside, low down: the XT60E-F turned on its side with 2 screws and their nuts inside, and the switch snapped in.

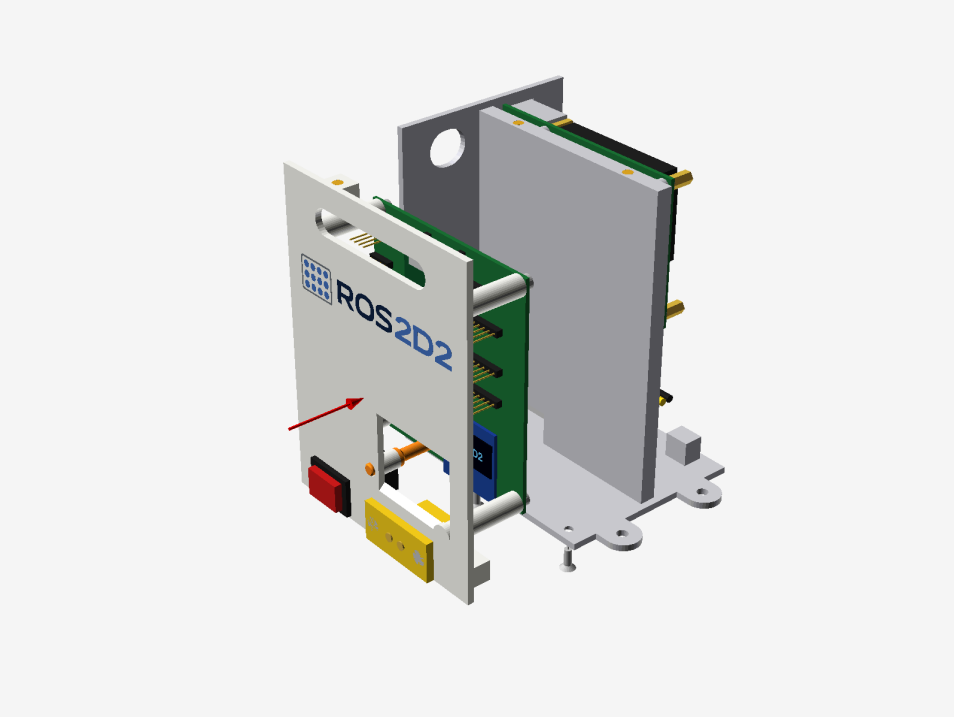
5 Push rod and Pico board



4× M3 × 6

Slide the rod into its guide from inside and screw the board onto the 4 columns, parts towards the wall. Tighten the 5 V in terminal J1, plug the angled micro-USB into the Pico and connect the sensors: their cables leave through the slot at the top.

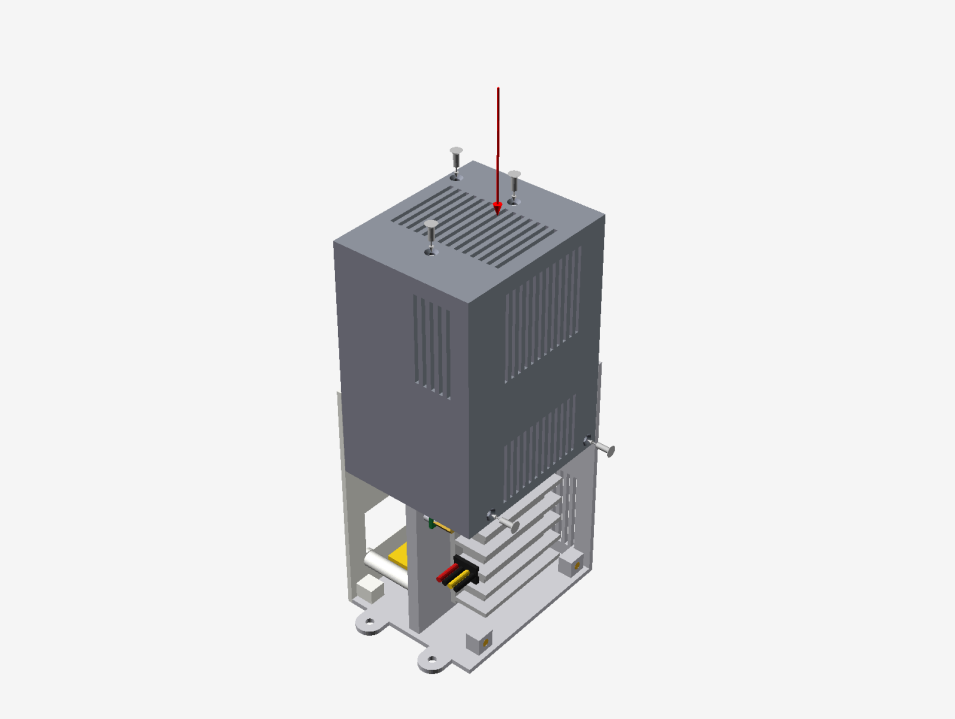
6 Main face onto the base



2× M3 × 8 countersunk

Run the 24 V wires (XT60 → switch → converter) and the 5 V to terminal J1 through the gap at the bottom of the divider. Screw the face on from under the floor.

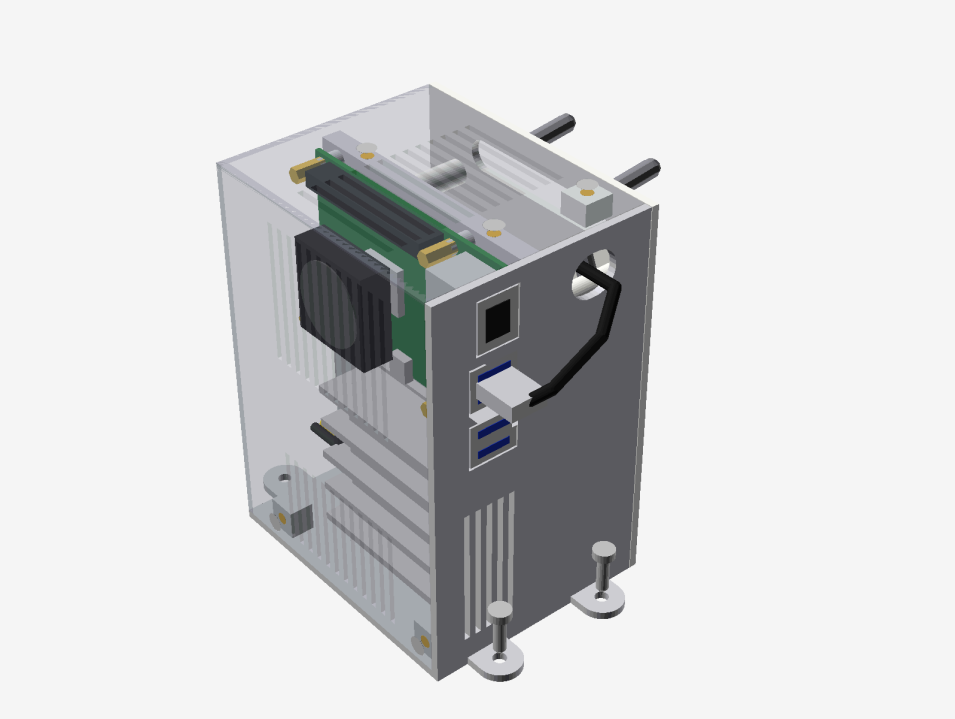
7 Shell



5× M3 × 8 countersunk

Lower it from above and fit the 3 roof screws and the 2 side screws.

8 USB and onto the R2



4× M4 × 12

Bring the micro-USB out of the notch in the back into a Pi USB port. Screw the box to the R2 through its 4 ears.